

Mehrdad (Mer) Khani

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PROFESSIONAL SUMMARY

Biotechnologist with 6+ years of pharmaceutical experience in **GMP** environments, including ~3.5 years in recombinant protein expression and purification across **mammalian** (CHO, HEK), insect (Sf9), and bacterial (E. coli) systems. Hands-on with **FPLC**-based multistep chromatography (Protein A, IEX, HIC, SEC) on ÄKTA systems, cell culture maintenance, and **downstream purification** workflows (TFF, UF/DF, nanofiltration). Experienced in buffer preparation, SOP and lab notebook documentation, and cross-functional collaboration with senior scientists across R&D and production teams.

PROFESSIONAL EXPERIENCE

API Production Technician | Dec 2023 – May 2026

Eurofins CDMO Alphora Inc. | Mississauga, ON, Canada

- Execute API manufacturing operations across **5 production suites**, handling batch scales from **5 L (kilolab) to 400 L (pilot plant)** across **~15 batch types**; operate glass reactors, vacuum dryers, mills, and filtration units per **BPRs** and **cGMP standards**, working on a **continental shift schedule**.
- Handle **Class 2 organic solvents** (e.g., THF), strong acids/bases, and high-potency intermediates using **full PPE up to Level P7** (Tyvek with supplied air); prepare scrubbers, execute **CIP procedures**, and carry out safe disposal of API residues with proper waste labeling and segregation.
- Perform **in-process sampling**, coordinate sample submission to QC, and execute **batch reconciliation calculations** with graphical comparison to historical batches; maintain batch timing data to support cycle-time analysis and warehouse reporting.
- Troubleshoot in-line equipment issues (e.g., **HPLC pump disassembly, pumps, scales**) to maintain production continuity; perform routine **calibration and maintenance** per documented SOPs.
- Act as **co-lead** in the absence of the supervisor; coordinate daily with **Tech Transfer, QA, QC, maintenance, planning, and warehouse** teams; train new staff on procedures, equipment, GMP, and workplace safety standards.
- Support recurring **internal and external audits** (including Japanese international and **FDA inspections**) with full **HSE and GDP compliance**; participate in **RCA meetings** for deviations and near-misses with preventive recommendations, and propose **BPR content improvements** where discrepancies between documented and executed procedures were identified.

Quality Assurance Specialist – Process Validation | Mar 2023 – Jul 2023

CinnaGen Biosimilar Company | Tehran, Iran

- Supported the preparation and execution of **Process Validation Protocols (PVPs)** and **Validation Reports (PVRs)** for new and existing biopharmaceutical manufacturing processes, in alignment with **Annex 15** (Qualification and Validation) and EU GMP requirements.
- Supported and executed **Cleaning Validation** studies, including swab/rinse sampling design, **MACO calculations, recovery studies, clean/dirty hold-time studies**, and risk-based documentation in line with **Annex 1** requirements.
- Reviewed and supported **Equipment Qualification (IQ/OQ/PQ)** protocols; applied risk-based methodologies (**FMEA**) to prioritize critical process parameters and validation activities.
- Provided in-process **QA oversight during aseptic filling operations**, monitoring **Grade A/B** environment compliance and adherence to **Annex 1** requirements.
- Revised and maintained **SOPs** and controlled documents through annual review cycles; supported investigations of process deviations, **OOS/OOT events**, and **CAPA** implementation in coordination with **manufacturing, QC, and engineering** teams.

Downstream Purification Specialist – R&D | Aug 2021 – Mar 2023

CinnaGen Biosimilar Company | Karaj, Iran

- Executed downstream purification workflows for the **adalimumab biosimilar** across multistep chromatography (**Protein A affinity, IEX, SEC, HIC**) on **ÄKTA systems**, following established R&D protocols.
- Prepared **buffers** and **culture media** per defined recipes; documented observations and method notes (e.g., dissolution behavior, deviations from expected results) to support iterative protocol refinement by the development team.
- Executed **TFF, depth filtration, and nanofiltration experiments** as designed by the development team; recorded process data and reported results for analysis.
- Performed **SDS-PAGE** to verify protein identity and purity during purification runs; reviewed and interpreted results in coordination with senior scientists.
- Served as a **liaison between R&D and Production, QC, QA, Tech Transfer, and Warehouse** departments; conducted routine site visits to production areas to support technical alignment and process transfer.
- Prepared **technical reports and PowerPoint presentations** summarizing experimental work and outcomes; trained production staff on processes, equipment operation, and underlying biology/chemistry principles.

Downstream Purification Technologist – Production | Oct 2019 – Aug 2021

CinnaGen Biosimilar Company | Karaj, Iran

- Executed GMP-compliant purification for recombinant proteins and mAbs (**adalimumab, eculizumab**, interferon β -1a, GLP-1 analogs, human growth hormone) in **BSL-2/3 cleanroom** environments; contributed to large-scale manufacturing of **1 million doses** of the **SPIKOGEN® COVID-19 vaccine**.
- Supported **upstream operations** across bioreactor scales (**5 L to 4500 L**), including cell culture, cell counting, and harvest; operated **disc-stack centrifuges**, **high-pressure homogenizers** for E. coli cell disruption, and **depth filtration** for bioburden reduction.
- Ran multistep chromatography trains (**Protein A affinity, IEX, SEC, HIC**) on **ÄKTA Pilot/Process** and **NOVASEP HPLC**; performed **viral inactivation** (low-pH hold), **viral filtration** (nanofiltration), **UF/DF (TFF)**, **final filtration**, and **bulk preparation** per validated SOPs.
- Documented all runs in **UNICORN, BPRs, and SCADA/iFIX**; performed real-time in-process testing (pH, conductivity, UV absorbance), **yield and mass balance calculations**, **QC sample collection**, and **environmental monitoring** (viable/non-viable particles, surface contact plates).
- Performed **CIP/SIP, column packing and qualification** (plate height, asymmetry), production-scale buffer/media preparation, and **aseptic operations**; troubleshooted production-scale equipment, **trained new production staff**, coordinated cross-functionally with **QC, QA, and Tech Transfer**, and worked a **continental shift schedule**.

EDUCATION

Bachelor of Science in Biotechnology | 2015 – 2019 | *University of Amol, Iran* | *WES Certified*

TECHNICAL SKILLS & SYSTEMS

Protein Purification & Method Development: ÄKTA Pilot/Process (UNICORN), NOVASEP HPLC, Protein A affinity, IEX, HIC, SEC; buffer design, elution optimization, resin screening

Analytical Methods: SDS-PAGE, UV/Vis spectroscopy, in-process testing (pH, conductivity, UV absorbance)

Filtration Workflows: TFF/UF-DF, depth filtration, viral filtration (nanofiltration), filter integrity testing

Manufacturing & Equipment: Disc-stack centrifuges, high-pressure homogenizers, CIP/SIP skids, vacuum dryers, reactors, mills

Cell Platforms: Mammalian (CHO, HEK) cell culture; E. coli fermentation harvest, inclusion body isolation, solubilization & refolding

Documentation & Compliance: BPRs, SOPs, technical reports, LIMS, SCADA/iFIX, GMP (EU GMP, ICH Q7, cGMP), deviation/CAPA management

Software: MS Office Suite, Adobe Acrobat, UNICORN, LIMS, Minitab